# PHYSICS

1) When two identical charges Q placed at separation r, then force between them is F. If one charge increased by 2C then force becomes 4F. What is the value of Q :-

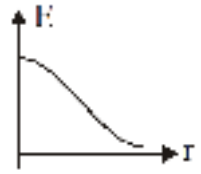
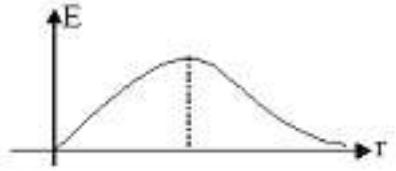
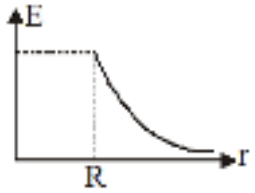
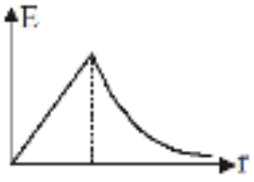
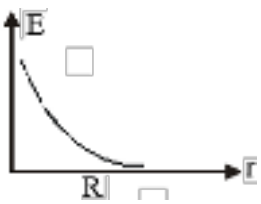
(1) $\frac{3}{2}C$

(2) 2C

(3) $\frac{2}{3}C$

(4) 4C

2) Column II corresponds to the graph of magnitude of electric field versus distance from centre of charge distribution in Column I. Match column-I with column-II.

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| (A) | Along axis of ring | (P) |   |
| (B) | Uniformly charged solid sphere | (Q) |   |
| (C) | Uniformly charged spherical shell | (R) |   |
| (D) | Combination of charge +Q and –Q at the perpendicular bisector | (S) |   |
| | | (T) |   |

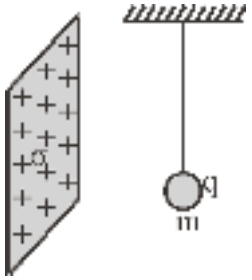
(1) (A)→(Q); (B)→(S); (C)→(R); (D)→(P)

(2) (A)→(S); (B)→(R); (C)→(P); (D)→(S)

(3) (A)→(R); (B)→(P); (C)→(Q); (D)→(R)

(4) (A)→(P); (B)→(R); (C)→(S); (D)→(Q)

3) A small charged particle of mass m and q is suspended by an insulated thread infront of a very large charged sheet of uniform surface density of charge σ. The angle made by the thread with the vertical in equilibrium is :-



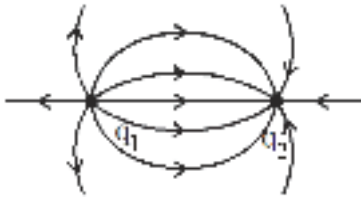
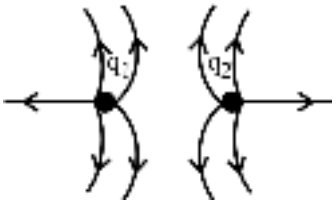
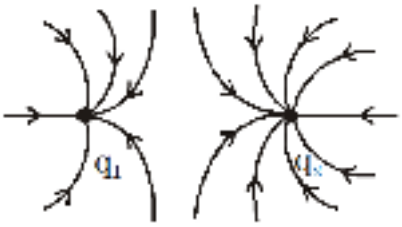
(1) $\tan^{-1}\left(\frac{\sigma q}{2\varepsilon_0 mg}\right)$

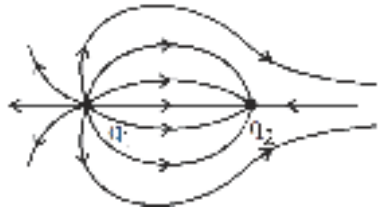
(2) $\tan^{-1}\left(\frac{\sigma}{q\varepsilon_0 mg}\right)$

(3) $\tan^{-1}\left(\frac{q}{2\sigma\varepsilon_0 mg}\right)$

(4) Zero

4) List–I shows the electric field lines diagram for two point charges $q_1$ (left) and $q_2$ (right). List–II has the possible information about the charges $q_1$ and $q_2$. Match the correct combination of charges with their field line diagram.

| List-I | | List-II | |
|---|---|---|---|
| (P) |  | (1) | $\lvert q_1\rvert > \lvert q_2\rvert$ |
| (Q) |  | (2) | $q_1 > 0$, $q_2 < 0$ |
| (R) |  | (3) | $q_1 > 0$, $q_2 > 0$ |

| | | | |
|---|---|---|---|
| (S) |  | (4) | $q_1 < 0$, $q_2 < 0$ |
| | | (5) | $\|q_1\| < \|q_2\|$ |

(1) P → 1; Q → 3; R → 3, 5; S → 3, 4

(2) P → 1, 2; Q → 3, 4; R → 4, 5; S → 1, 2

(3) P → 2; Q → 3; R → 4, 5; S → 1, 2

(4) P → 3, 4; Q → 1, 2; R → 4, 5; S → 1, 2

5) Five positive equal charges are placed at vertices of a regular hexagon and net electric field at the centre is $E_1$. A negative charge having equal magnitude is placed at sixth vertex and then net electric field is $E_2$. Find $\frac{E_2}{E_1}$.

(1) 2

(2) 1

(3) 3

(4) None of these

6) Two fixed charges A and B of 5μC each are separated by a distance of 6m. C is the mid point of the line joining A and B. A charge 'Q' of –5μC is shot perpendicular to the line joining A and B from C with a kinetic energy of 0.06J. The charge 'Q' comes to rest at a point D. The distance CD is :-

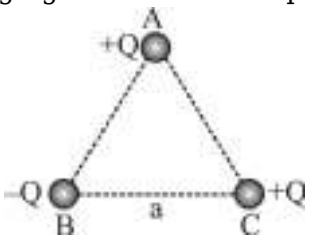
(1) 3m

(2) $\sqrt{3}$ m

(3) $3\sqrt{3}$ m

(4) 4m

7) Three charges are placed at the vertices of an equilateral triangle of side 'a' as shown in the following figure. The force experienced by the charge placed at the vertex A in a direction normal to BC is :-




(1) $Q^2/(4\pi\varepsilon_0 a^2)$

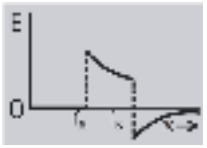
(2) $-Q^2/(4\pi\varepsilon_0 a^2)$

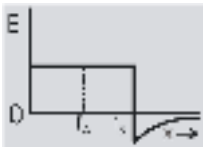
(3) Zero

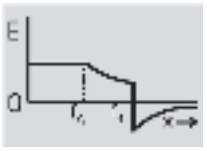
(4) $Q^2/(2\pi\varepsilon_0 a^2)$

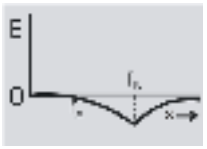
8) Two concentric conducting thin spherical shells A, and B having radii $r_A$ and $r_B$ ($r_B > r_A$) are

charged with charges $Q_A$ and $-Q_B$ ($|Q_B| > |Q_A|$). The electric field along a line, (passing through the centre) varies with distance x from the centre as :-

(1) 


(2) 

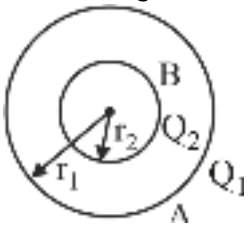

(3) 


(4) 


9) If a body is charged by rubbing it, its weight :-

(1) always decreases slightly
(2) always increases slightly
(3) may increase/decrease slightly
(4) remains precisely the same

10) Charges on spherical shell A & B are $Q_1$ and $Q_2$ of radius $r_1$ and $r_2$. The spheres are joined by a conducting wire and then separated out, the surface charge density of A and B are :




(1) $\frac{Q_1}{4\pi r_1^2}, \frac{Q_2}{4\pi r_2^2}$

(2) $\text{Zero}, \frac{Q_2}{4\pi r_2^2}$

(3) $\frac{Q_1 + Q_2}{4\pi r_1^2}, \text{Zero}$

(4) $\text{Zero}, \frac{Q_1 + Q_2}{4\pi r_1^2}$

11) A glass rod rubbed with silk is used to charge an air filled gold leaf electroscope then leaves will diverge. When the electroscope is exposed to X-rays for a short period. Then

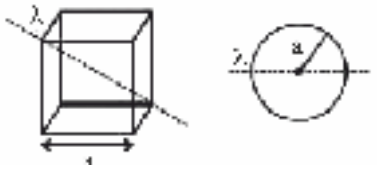
(1) The divergence of leaves will not be affected

(2) The leaves will diverge farther

(3) The leaves will collapse

(4) The leaves will melt

12) A linear charge having linear charge density λ, penetrates a cube diagonally so as to have maximum flux and then it penetrates a sphere diametrically as shown. What will be the ratio of flux coming out of cube and sphere ?



(1) $\frac{1}{2}$

(2) $\frac{2}{\sqrt{3}}$

(3) $\frac{\sqrt{3}}{2}$

(4) $\frac{1}{1}$

13) On the axis of uniformly charged ring, of radius R, distance between two maxima of electric field is :

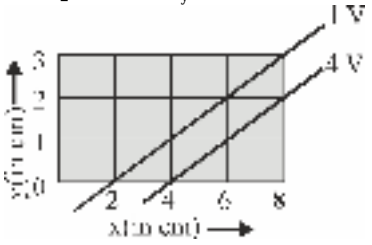
(1) $\frac{R}{\sqrt{2}}$

(2) $R\sqrt{2}$

(3) 2R

(4) $\frac{R}{2}$

14) Figure shows two equipotential lines in xy-plane for an electric field. The x-component $E_x$ and y-component $E_y$ of the field in the electric space between these equipotential lines are, respectively :




(1) + 150 V/m, –300 V/m

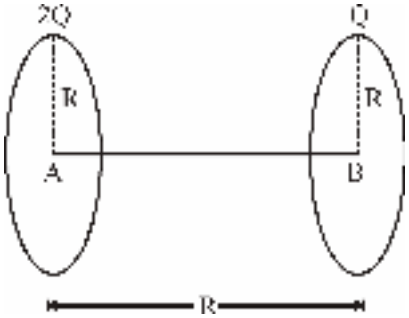
(2) –150 V/m, + 300 V/m

(3) + 200 V/m, 100 V/m

(4) –200 V/m, –100 V/m

15)

Which of the following value of electric potential is correct :




(1) $V_A = \frac{Q(2\sqrt{2}-1)}{4\pi\in_0\sqrt{2}R}$

(2) $V_B = \frac{Q(\sqrt{2}-1)}{4\pi\in_0 R}$

(3) $V_A = 0$

(4) None of these

16) An electric dipole of dipole moment $\vec{p} = \left(\hat{i}+2\hat{j}\right) C-m$ is kept in uniform electric field $\vec{E} = \left(\hat{i}+2\hat{j}+3\hat{k}\right) V/m$. Torque acting on dipole will be :-

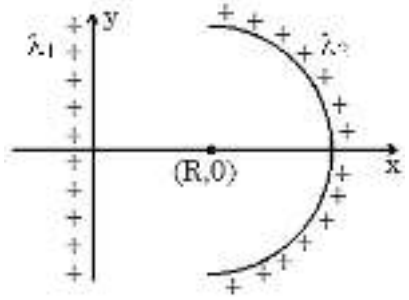
(1) $\left(6\hat{i}-3\hat{j}\right) N-m$

(2) $\left(\hat{i}+3\hat{k}\right) N-m$

(3) $\left(3\hat{i}+2\hat{j}+\hat{k}\right) N-m$

(4) zero

17) A uniformly charged infinite wire is placed along 'y' axis and passing through the origin having linear charge density '$\lambda_1$'. A semicircle wire having radius R and linear charge density '$\lambda_2$' centred at (R, 0) is placed as shown. Find the ratio of $\frac{\lambda_1}{\lambda_2}$, if electric field at (R, 0) is zero.



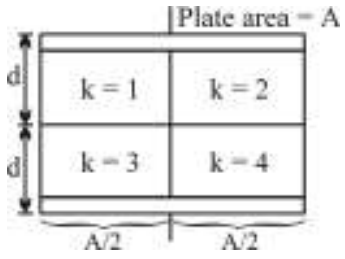

(1) $\frac{\lambda_1}{\lambda_2} = 1$

(2) $\frac{\lambda_2}{\lambda_1} = 2$

(3) $\frac{\lambda_2}{\lambda_1} = \frac{1}{2}$

(4) $\frac{\lambda_1}{\lambda_2} = \sqrt{2}$

18) Find the capacitance of the system shown in figure.



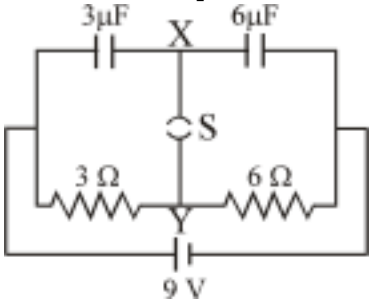

(1) $\frac{4\varepsilon_0 A}{7d}$

(2) $\frac{25}{24}\frac{\varepsilon_0 A}{d}$

(3) $\frac{24}{25}\frac{\varepsilon_0 A}{d}$

(4) $\frac{25}{12}\frac{\varepsilon_0 A}{d}$

19) A circuit is connected as shown in the figure with the switch S open. When the switch is closed, the total amount of charge that flows from Y to X is :-



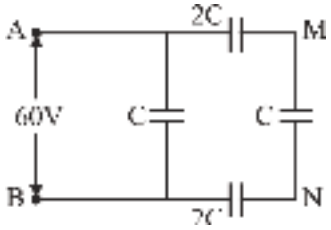

(1) 0

(2) 54 μC

(3) 27 μC

(4) 81 μC

20) In the circuit shown, a potential difference of 60V is applied across AB. The potential difference between the point M and N is :-



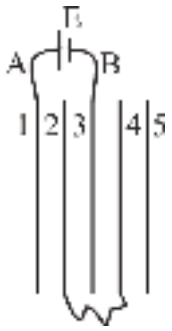

(1) 10 V

(2) 15 V

(3) 20 V

(4) 30 V

21) Five conducting parallel plates having area A and separation between them is d, are placed as shown in the figure. Plate number 2 and 4 are connected by a wire and between points A and B, a




cell of emf E is connected. The charge flown through the cell is :-

(1) $\frac{3}{4}\frac{\varepsilon_0 AE}{d}$

(2) $\frac{2}{3}\frac{\varepsilon_0 AE}{d}$

(3) $\frac{4\varepsilon_0 AE}{d}$

(4) $\frac{\varepsilon_0 AE}{2d}$

22) After charging a capacitor, battery is removed. Now by placing a dielectric slab between the plates :-

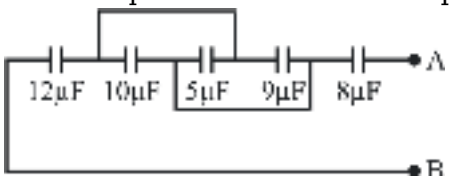
(1) The potential difference between the plates and the energy stored will decrease but the charge on plates will remain same

(2) The charge on the plates will decrease and the potential difference between the plates will increase

(3) The potential difference between the plates will increase and energy stored will decrease but the charge on the plates will remain same

(4) The potential difference, energy stored and the charge will remain unchanged.

23) Five capacitors together with their capacitances are shown in the adjoining figure. The potential difference between the points A and B is 60 volt. The equivalent capacitance between the point A and B and charge on capacitor 5μF will be respectively:-




(1) 44μF, 30μC

(2) 16μF, 150μC

(3) 15μF, 200μC

(4) 4μF, 50μC

24) A parallel plate capacitor, is made of two circular plates separated by a distance of 5mm and with a dielectric of dielectric constant 2.2 between them. When the electric field in the dielectric is $3 \times 10^4$ V/m, the charge density of the positive plate will be close to:

(1) $3 \times 10^{-7}$ C/m$^2$

(2) $3 \times 10^4$ C/m$^2$

(3) $6 \times 10^4$ C/m$^2$

(4) $6 \times 10^{-7}$ C/m$^2$

25) In a spherical condenser, radius of the outer sphere is R. The difference in the radii of outer and inner sphere is x. Its capacity is proportional to

(1) $\frac{xR}{(R-x)}$

(2) $\frac{x(R-x)}{R}$

(3) $\frac{R(R-x)}{x}$

(4) $\frac{R}{x}$

26) A galvanometer of resistance 100 Ω contains 100 division. It gives a deflection of one division on passing a current of $10^{-4}$ A. Find the resistance in ohms to be connected to it, so that it becomes a voltmeter of range 10V.

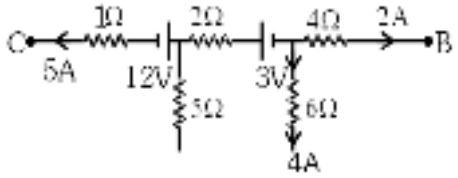
(1) $\frac{500}{9}\Omega$

(2) 500Ω

(3) $\frac{100}{9}\Omega$

(4) 900Ω

27) The figure shows part of certain circuit, find :




Potential difference $V_C - V_B$ :

(1) 6 V

(2) 10 V

(3) 4 V

(4) 8 V

28) Two electric bulbs whose resistance are in the ratio of 1 : 2, are connected in series to a constant voltage source. The power dissipated in them has the ratio :-

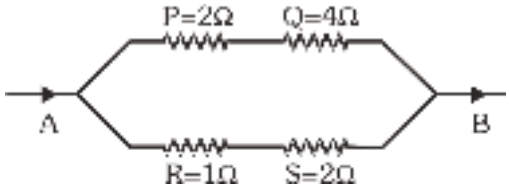
(1) 1 : 2

(2) 1 : 1

(3) 1 : 4

(4) 2 : 1

29) Which of the four resistances generate the greatest amount of heat when a current flows from A



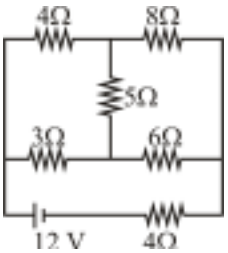

to B?

(1) P

(2) Q

(3) R

(4) S




30) In the given figure the ratio of current in 8Ω and 3Ω will be :-

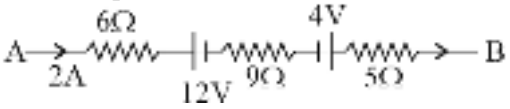
(1) $\frac{8}{3}$

(2) $\frac{3}{8}$

(3) $\frac{4}{3}$

(4) $\frac{3}{4}$

31) The potential difference between A and B in the following figure is:




(1) 24 V

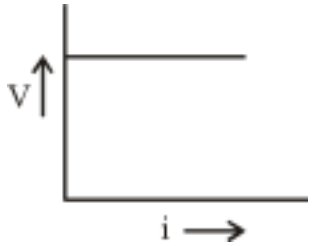
(2) 14 V

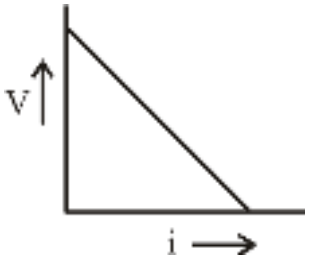
(3) 32 V

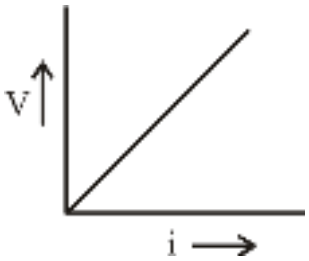
(4) 48 V

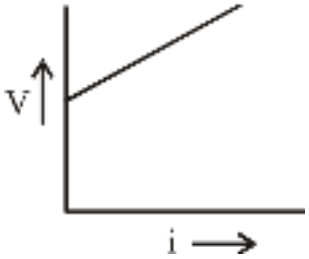
32) Graphs are drawn between current 'i' taken from a cell and the terminal voltage V of the cell. Which one is correct :-

(1)



(2) 


(3) 


(4) 

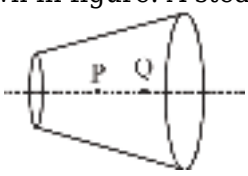

33) An unknown resistance $R_1$ is connected in series with a resistance of 10 Ω. This combination is connected to one gap of a metre bridge, while a resistance $R_2$ is connected in the other gap. The balance point is at 50 cm. Now when the 10 Ω resistance is removed, the balance point shifts to 40 cm. The value of $R_1$ is :-

(1) 10 Ω
(2) 20 Ω
(3) 40 Ω
(4) 60 Ω

34) **Assertion :** Manganin and constantan are widely used in standard resistors. **Reason :** Manganin and constantan resistance value would change very little with temperatures.

(1) Both assertion & reason are true & reason is a correct explanation of the assertion.
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.
(3) Assertion is true but reason is false.
(4) Both assertion & reason are false.

35) A wire has non-uniform cross-section as shown in figure. A steady current flows through it and electric field at points P and Q is $E_P$ and $E_Q$, then




(1) $E_P < E_Q$
(2) $E_P = E_Q$
(3) $E_P > E_Q$

(4) data insufficient

36)

In the absence of applied potential, the electric current flowing through a metallic wire is zero because –

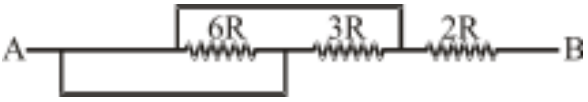
(1) The average velocity of electron is zero.

(2) The electrons are drifted in random direction with a speed of order of $10^{-2}$ cm/s.

(3) The electrons move in random direction with a speed of the order close to that of velocity of light.

(4) Electron and ions move in opposite direction.

37) Equivalent resistance between A and B is :-



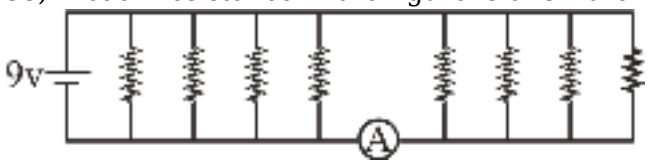

(1) 5R

(2) 2R

(3) $\frac{6R}{5}$

(4) R

38) If each resistance in the figure is of 3Ω then reading of the ammeter is :-



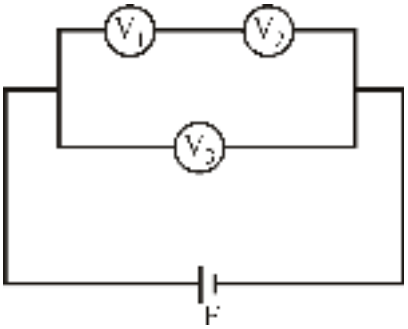

(1) 12 A

(2) 15 A

(3) 9 A

(4) 6 A

39) Compare readings of identical voltmeters :-



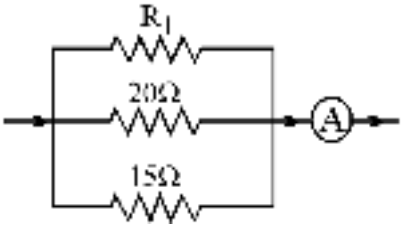

(1) $(V_1 = V_3) > V_2$

(2) $(V_1 = V_2) > V_3$

(3) $(V_1 = V_2) < V_3$

(4) $V_1 = V_2 = V_3$

40) In the given circuit, the current flowing through the resistance 20 ohms is 0.3 ampere while the ammeter reads 0.8 ampere. What is the value of $R_1$?




(1) 30 ohms

(2) 40 ohms

(3) 50 ohms

(4) 60 ohms

41) Potential difference between centre and the surface of sphere of radius R and uniform volume charge density ρ will be :

(1) $\frac{\rho R^2}{6\epsilon_0}$

(2) $\frac{\rho R^2}{4\epsilon_0}$

(3) 0

(4) $\frac{\rho R^2}{2\epsilon_0}$

42) Five cells each of e.m.f. E and internal resistance r are connected in series. If due to oversight, one cell is connected wrongly, then the equivalent e.m.f. and equivalent internal resistance of the combination is:

(1) 5 E, 5r

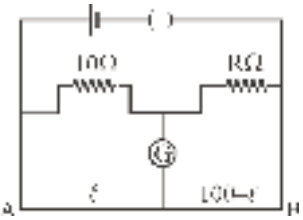
(2) 2 E, 5r

(3) 3 E, 5r

(4) 4 E, 3r

43) The resistance in the two arms of the meter bridge are 10Ω and RΩ respectively and balance point is at '□'. When another same resistance R is added in series of R, new balance point is at $\frac{3}{4}$□.

The resistance R is :-



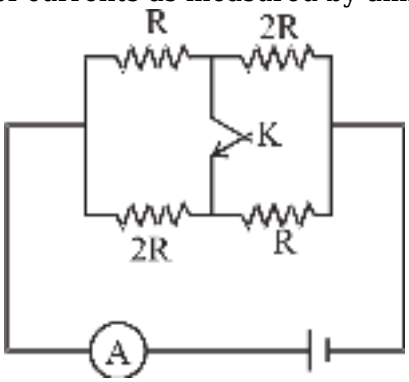

(1) 5 Ω

(2) 10 Ω

(3) 15 Ω

(4) 20 Ω

44) Find the ratio of currents as measured by ammeter in two cases when the key is open and when the key is closed :-




(1) $\frac{9}{8}$

(2) $\frac{10}{11}$

(3) $\frac{8}{9}$

(4) None of the above

45) Which of these is a false statement ?

(1) The wheatstone bridge is most sensitive when all the four resistances are of the same order

(2) In a balanced wheatstone bridge, interchanging the positions of galvanometer and cell affects the balance of the bridge

(3) Kirchhoff's first law (for currents meeting at a junction in an electric circuit) expresses the conservation of charge

(4) The rheostat can be used as a potential divider

## CHEMISTRY

1) Suitable indicator for the acid-base titration of HCOOH against KOH is.

(1) Methyl orange

(2) Methyl red

(3) Phenolphthalein

(4) Thymol Blue

2) 200 $cm^3$ of 0.1M $H_2SO_4$ solution is mixed with 150 $cm^3$ of 0.2M KOH solution. Find the value of evolved heat. [Enthalpy of neutralization is –57.2 kJ/equivalent]

(1) 1.7 kJ

(2) 2.7 kJ

(3) 9.2 kJ

(4) $1.9 \times 10^3$ kJ

3) Bredig's arc method cannot be used for the preparation of colloidal sol of :-

(1) Copper (Cu)

(2) Gold (Au)

(3) Silver (Ag)

(4) Sodium (Na)

4) Molar mass of Mohr's salt is 392. In Mohr's salt iron is oxidised by $KMnO_4$ in acidic medium. Equivalent weight of Mohr's salt is :

(1) 392

(2) 31.6

(3) 278

(4) 156

5) In the titration of oxalic acid and $KMnO_4$, which acid is added ?

(1) $HNO_3$

(2) HCl

(3) $H_2SO_4$

(4) Any of the above

6) **Statement-I :** Dilute $H_2SO_4$ is added in preparing standard solution of ferrous ammonium sulphate to prevent the hydrolysis of ferrous salt.
**Statement-II :** $KMnO_4$ is reducing agent and Mohr's salt is oxidising agent in titration of both Mohr's salt and $KMnO_4$.

(1) Statement-I is true and Statement-II is wrong

(2) Statement-I is wrong and Statement-II is true

(3) Both statements are true

(4) Both statements are wrong

7) For a reaction of Iodide ion with hydrogen peroxide at room temperature, what will be the order of reaction concerning iodide ion :

(1) $I^{st}$ order

(2) Zero order

(3) $II^{nd}$ order

(4) $III^{rd}$ order

8) A 0.20M KOH solution is electrolysed for 1.5 h using a current of 8.00 A. How many moles of product obtained at anode ? (At. wt. of potassium, hydrogen and oxygen = 39, 1, 16 respectively)

(1) $2.24 \times 10^{-2}$

(2) 0.224

(3) 0.112

(4) 0.48

9) What is colour of starch iodine complex formed in kinetic study of reaction between $I^-$ & $H_2O_2$ at room temperature ?

(1) Yellow
(2) Blue
(3) Green
(4) Red

10) If a reaction has a velocity constant equal to $10^{-2}$ $s^{-1}$ then the time will it take for 20 g of the reactant to reduce by 15g.

(1) 27.67 s
(2) 346.5 s
(3) 693.0 s
(4) 138.6 s

11) Collision theory is satisfactory for-

(1) First order reaction
(2) Zero order reaction
(3) Bimolecular elementary reactions
(4) Any order of reaction

12) **Assertion (A) :** If the activation energy is zero temperature will have no effect on the rate constant.
**Reason (R) :** Lower the activation energy faster the reaction.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion is incorrect but Reason is correct.

13) For an endothermic reaction ΔH (KJ $mol^{-1}$) is enthalpy change of reaction the minimum value of the energy of activation will be-

(1) Zero
(2) More than ΔH
(3) Equal to ΔH
(4) Less than ΔH

14) For equimolar solutions of urea, $BaCl_2$ and $AlCl_3$ the correct order of boiling point is-

(1) urea > $BaCl_2$ > $AlCl_3$
(2) urea > $AlCl_3$ > $BaCl_2$
(3) $BaCl_2$ > $AlCl_3$ > urea
(4) $AlCl_3$ > $BaCl_2$ > urea

15) Value of Henry's Constant for $N_2$ and $O_2$

(1) Increase with increase in temperature
(2) Decrease with increase in temperature
(3) Remains constant with increase in temperature
(4) First increase then decrease with increase in temperature

16) **Assertion (A) :** Isotonic solutions must have the same molar concentrations.
**Reason (R) :** Solutions which have the same osmotic pressure are known as isotonic solutions.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion is incorrect but Reason is correct.

17) The temperature at which the vapour pressure of a liquid become equal to the atmospheric pressure is known as-

(1) Freezing point
(2) Boiling point
(3) Absolute temperature
(4) Not such temperature possible

18) The specific conductance of a 0.1N KCl solution at 296 K is 0.012 $ohm^{-1}cm^{-1}$ . The resistance of cell containing the solution at the same temperature was found to be 55 ohm the cell constant is (in $cm^{-1}$)

(1) 0.66
(2) 0.918
(3) 1.12
(4) 0.142

19) The EMF of a galvanic cell if $E^{o}_{M/M^{n+}}$ for a metal M at anode is –0.34 V and $E^{o}_{N^{n+}/N}$ for another metal N at cathode is 0.81 V is-

(1) –0.47 V
(2) 1.15 V
(3) +0.47 V
(4) –1.15 V

20) Value of ionisation constant of a weak acid in terms of $\wedge_m$ and $\wedge_m^{\infty}$ is

(1) $K_a = \frac{C\Lambda_m^{\infty}}{\Lambda_m - \Lambda_m^{\infty}}$

(2) $K_a = \frac{C(\Lambda_m^{\infty})^2}{\Lambda_m^{\infty}(\Lambda_m^{\infty} - \Lambda_m)}$

(3) $K_a = \frac{C\Lambda_m^2}{\Lambda_m^\infty(\Lambda_m^\infty - \Lambda_m)}$

(4) $K_a = \frac{\Lambda_m}{\Lambda_m^\infty}$

21) How many coulombs are provided by a current of 0.01 mA in a battery which is used for 10 hours?

(1) 0.36

(2) 0.10

(3) 10

(4) 6600

22) The equilibrium constant for the reaction
$2A_2(s) + 3B^{2+}(aq.) \rightarrow 2A_2^{3+}(aq.) + 3B(s)$
is $10^{24}$ then $E^o$ for the cell at 298 K is-

(1) 0.084 V

(2) 0.2364 V

(3) 0.0788 V

(4) –0.2364 V

23) For the reaction $A_2 + 2B \rightarrow 2AB$

| $[A_2]$ M | [B] M | $\frac{-d\,[A_2]}{dt}$ $Ms^{-1}$ |
|---|---|---|
| 0.2 | 0.3 | $5 \times 10^{-2}$ |
| 0.2 | 0.6 | 0.2 |
| 0.8 | 0.9 | 1.8 |

then order of reaction with respect to B and $A_2$ respectively are-

(1) $0, \frac{5}{2}$

(2) $2, \frac{1}{2}$

(3) 4, 2

(4) 2, 1

24) Match the column-I to column-II.

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| (P) | Mathematical expression for rate of reaction | (A) | Rate constant |
| (Q) | Rate of reaction for zero order reaction is equal to | (B) | Rate law |
| (R) | Unit of rate constant for zero order reaction is same as that of | (C) | Order of slow step |

| | | | |
|---|---|---|---|
| (S) | Order of a complex reaction is determined by | (D) | Rate of reaction |

Correct is -

| | (P) | (Q) | (R) | (S) |
|---|---|---|---|---|
| (1) | (B) | (A) | (D) | (C) |
| (2) | (B) | (A) | (C) | (D) |
| (3) | (A) | (B) | (D) | (C) |
| (4) | (A) | (B) | (C) | (D) |

(1) 1

(2) 2

(3) 3

(4) 4

25) In presence of catalyst $E_a$ is lowered by 2kcal at 27°C, rate will be-

(1) 20 times

(2) 14 times

(3) 28 times

(4) 2 times

26) If time 't' is required for a radioactive substance to become one third of its initial amount, what fraction would be left after $\frac{t}{2}$ ?

(1) $\frac{1}{\sqrt{3}}$

(2) $\frac{1}{3}$

(3) $\frac{1}{2}$

(4) $\sqrt{\frac{2}{3}}$

27) A solution containing 3g $L^{-1}$ of a non electrolyte solute exerts same osmotic pressure as that of $5 \times 10^{-2}$ M aluminium phosphate solution, molecular weight of the non electrolyte solute is-

(1) 180

(2) 50

(3) 120

(4) 30

28) At 300 K, 40 mL of ozone gas dissolve in 100 g of water at 1 atm, find the mass of same gas dissolved in 400g water at 4 atm and 27°C temperature.

(1) 1.24 g

(2) 0.1 g

(3) 0.48 g

(4) 4.8 g

29) Match the column-I to column-II.

| **Column-I** | | **Column-II** | |
|---|---|---|---|
| (P) | Soda water | (A) | a solution of gas in solid |
| (Q) | Sugar solution | (B) | a solution of gas in gas |
| (R) | German silver | (C) | a solution of solid in liquid |
| (S) | Air | (D) | a solution of solid in solid |
| (T) | Hydrogen gas in palladium | (E) | a solution of gas in liquid |
| | | (F) | a solution of liquid in solid |

Correct is-

| | **(P)** | **(Q)** | **(R)** | **(S)** | **(T)** |
|---|---|---|---|---|---|
| (1) | (C) | (F) | (A) | (D) | (B) |
| (2) | (E) | (C) | (D) | (B) | (A) |
| (3) | (B) | (D) | (C) | (A) | (E) |
| (4) | (E) | (B) | (C) | (D) | (A) |

(1) 1

(2) 2

(3) 3

(4) 4

30) If at a certain temperature the vapour pressure of pure water is 25 mm of Hg and vapour pressure of very dilute aq. urea solution is 24.5 mm of Hg, molality of solution is-

(1) 0.06

(2) 0.02

(3) 1.20

(4) 1.11

31) Calculate $E^{o}$ of the half cell reaction $Cu^{+} + e^{-} \rightarrow Cu(s)$
if $Cu^{2+} + e^{-} \rightarrow Cu^{+}$, $E^{o} = 0.15$ V
and $Cu^{2+} + 2e^{-} \rightarrow Cu(s)$, $E^{o} = 0.34$ V

(1) –0.19 V

(2) 0.26 V

(3) 0.53 V

(4) 0.83 V

32) The molar conductivity of 0.25 M formic acid is 40.1 $Scm^{2}\ mol^{-1}$, then degree of dissociation of

acid is
[$(\Lambda_m^o)_{H^+}$ = 349.6 S $cm^2$ $mol^{-1}$ and $(\Lambda_m^o)_{HCOO^-}$ = 54.6 $Scm^2$ $mol^{-1}$]

(1) 11.113

(2) 9.097

(3) 0.718

(4) 0.099

33) $E^o_{cell} = 1.1V$ for Daniell cell which of the following expression is correct for equilibrium in this cell?

(1) $\log K_c = 1.1$

(2) $\log K_c = \frac{3.4}{0.059}$

(3) $\frac{2.303RT}{2F} \log K_c = 1.1$

(4) $1.1 = K_c$

34) In which of the following cell, reactants are continuously supplied from external source?

(1) Lead storage battery

(2) Fuel cell

(3) Lithium battery

(4) Dry cell

35) A reaction was started with a decimolar solution of the reactant, 8 min 20 sec later it's concentration was found to be $\frac{M}{100}$ rate of reaction is (in $Ms^{-1}$)

(1) $1.8 \times 10^{-3}$

(2) $1.8 \times 10^{-4}$

(3) $3.6 \times 10^{-4}$

(4) $3.6 \times 10^{-3}$

36) $P(g) \xrightarrow{\Delta} A(g) + B(g) + C(g)$ follow $I^{st}$ order with half life is 69.3 s at 500°C in a closed vessel with initial pressure of P gas is 0.4 atm, total pressure of system after 230 sec is :

(1) 1.15 atm

(2) 1.22 atm

(3) 1.32 atm

(4) 1.12 atm

37) When 25g $Na_2SO_4$ (Mw = 142) is dissolved in $10^3$ kg of solution it's concentration in ppm is-

(1) 0.176

(2) 2.5

(3) 25

(4) $2.5 \times 10^4$

38) Density of aqueous solution of $FeCl_3$ (formula mass = 162) is 1.1 g $mL^{-1}$ and contain 20% by mass $FeCl_3$. Concentration of this solution is-

(1) $2.8 \times 10^{-2}$ M

(2) 1.36 M

(3) 1.27 M

(4) 1.47 M

39) Given the following reactions involving A, B, C, D
(P) $C + B^+ \rightarrow C^+ + B$
(Q) $A^+ + D \rightarrow$ No reaction
(R) $C^+ + A \rightarrow$ No reaction
(S) $D + B^+ \rightarrow D^+ + B$
Correct order of their decreasing ability as reducing agent-

(1) D > B > C > A

(2) A > C > D > B

(3) C > A > B > D

(4) C > A > D > B

40) **Statement-I :** The rate of a reaction normally increases 2 to 3 times for every 10° rise in temperature.
**Statement-II :** Increase in temperature increase the number of collisions.

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

41) The molecularity of a complex reaction $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$ is-

(1) 2

(2) 1

(3) 3

(4) has no meaning

42) If two substances P and Q have $P_P^o : P_Q^o = 1 : 2$ and mole ratio of Q to P in solution is 2 : 1 then mole fraction of P in vapour is-

(1) 0.33

(2) 0.52

(3) 0.20

(4) 0.25

43) **Statement-I :** Addition of ethylene glycol to water lowers the freezing point of water therefore used as antifreeze substance.
**Statement-II :** Ice melts faster at high altitude.

(1) Both Statement-I and Statement-II are true.
(2) Both Statement-I and Statement-II are false.
(3) Statement-I is true but Statement-II is false.
(4) Statement-I is false but Statement-II is true.

44) If $E^{o}_{Zn/Zn^{2+}}$ and $E^{o}_{Cl_2/Cl^-}$ are 0.76 V and 1.36 V respectively then EMF of the cell- $Zn_{(s)}|Zn^{2+}(1M)$ || $Cl^-(0.5\ M)\ |Cl_2(g)$ (1 atm), Pt is-

(1) 2.082
(2) 2.129
(3) 2.138
(4) –0.68

45) Equivalent conductance of saturated $BaSO_4$ is 400 $\Omega^{-1}\ cm^2eq^{-1}$ and specific conductance is $8 \times 10^{-5}\Omega^{-1}cm^2$. $K_{sp}$ of $BaSO_4$ is -

(1) $4 \times 10^{-8}\ M^2$
(2) $1 \times 10^{-8}\ M^2$
(3) $2 \times 10^{-4}\ M^2$
(4) $1 \times 10^{-4}\ M^2$

## BIOLOGY

1) Potato tuber is considered as stem because of :

(1) Presence of nodes and internodes
(2) Presence of photosynthetic leaves
(3) Absence of root hair and root cap
(4) Storage of food

2) Swollen placenta, Oblique septum, Pentamerous, Bicarpellary, Actinomorphic, Hypogynous. Above characters are related to :-

(1) Cruciferae family
(2) Liliaceae family
(3) Solanaceae family
(4) Fabaceae family

3) In which of the following flower both adhesion and cohesion of stamens are present ?

(1) Chinarose

(2) Mustard

(3) Datura

(4) Pea

4) Match the column-I with column-II :

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | Hypogynous | (i) | Rose |
| (B) | Perigynous | (ii) | Chilli |
| (C) | Epigynous | (iii) | Brinjal |
| (D) | Actinomorphic | (iv) | Apple |

(1) A-i, B-ii, C-iii, D-iv

(2) A-iv, B-iii, C-ii, D-i

(3) A-iii, B-ii, C-i, D-iv

(4) A-iii, B-i, C-iv, D-ii

5) Which of the following statements are correct ?
(A) Endosperm of castor is formed as a result of double fertilization.
(B) Outer covering of endosperm in maize separates the embryo by a proteinaceous layer.
(C) In the seeds of maize plumule and radicle are enclosed in sheath which are called coleorhiza and coleoptile respectively.
(D) Development of fruit without fertilization of ovary is known as parthenocarpy.

(1) A & B only

(2) A, B & D only

(3) A, B & C only

(4) A, B, C, D all

6) **Assertion :** Endosperm is edible in all drupe fruits.
**Reason :** The endosperm is stony hard in coconut.

(1) Assertion and Reason are correct and Reason is correct explanation of Assertion

(2) Both Assertion and Reason are correct but Reason is not correct explanation of Assertion

(3) Assertion is correct but Reason is incorrect

(4) Both Assertion and Reason are incorrect

7) Read the following statements and select the correct option given below :
**Statement-I :** A bud is present in the axil of petiole of simple leaves but not in the axil of petiole of compound leaves.
**Statement-II :** A bud is present in the axil of leaflets of the compound leaf.

(1) Statement-I and Statement-II both are correct

(2) Statement-I and Statement-II both are incorrect

(3) Only statement-I is correct

(4) Only Statement-II is incorrect

8) Match the following

| I | | II | |
|---|---|---|---|
| (i) | Mango | (a) | Fibrous mesocarp |
| (ii) | Castor | (b) | Non-endospermous seed |
| (iii) | Coconut | (c) | Endospermous seed |
| (iv) | Pea | (d) | Fleshy edible mesocarp |

(1) (i) a, (ii) b, (iii) c, (iv) d
(2) (i) c, (ii) d, (iii) a, (iv) b
(3) (i) d, (ii) c, (iii) a, (iv) b
(4) (i) b, (ii) a, (iii) d, (iv) c

9) Axile placentation is found in -

(1) Members of Liliaceae family
(2) Members of Solanaceae family
(3) Members of Fabaceae family
(4) (1) and (2) both

10) The pattern of arrangement of leaves on the stem or branch is called -

(1) Venation
(2) Phyllotaxy
(3) Aestivation
(4) Placentation

11) Examine the figure A–D given below and select the right option in which all the four figures A, B, C and D are identified correctly.

(A) 

(B) 

(C) 

(D) 

| | (A) | (B) | (C) | (D) |
|---|---|---|---|---|
| (1) | Racemose inflorescence | Hypogynous Flower | Twisted aestivation | Axila placentation |
| (2) | Cymose inflorescence | Hypogynous Flower | Twisted aestivation | Axila placentation |
| (3) | Racemose inflorescence | Perigynous Flower | Valvate aestivation | Marginal placentation |
| (4) | Cymose inflorescence | Epigynous Flower | Imbricate Aestivation | Free central placentation |

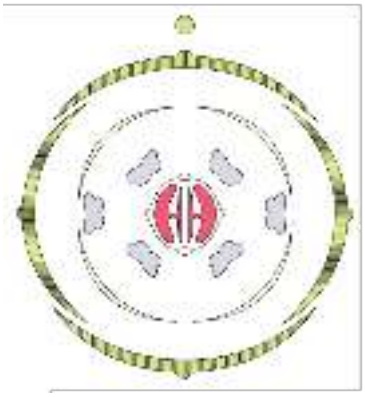
(1) 1

(2) 2

(3) 3

(4) 4

12) Identify the floral diagram shown below as well as related plant of its occurrence and select the right option for the two together ?



| Family | | Plant |
|---|---|---|
| (1) | Liliaceae | Onion |
| (2) | Brassicaceae | Mustard |
| (3) | Fabaceae | Pea |
| (4) | Solanaceae | *Petunia* |

(1) 1

(2) 2

(3) 3

(4) 4

13) Read all statements carefully and select correct statements :
(A) Pollen tube enters into ovule directly through micropyle known as porogamy
(B) Example of Apocarpous Gynoecium : *Papaver, Hibiscus*
(C) Syncarpous Gynoecium : *Papaver, Hibiscus*
(D) The arrangement of ovules within the ovary is known as placentation.

(1) A, C and D

(2) A, B and D

(3) Only B

(4) Only D

14) Radial symmetry is not found in ?

(1) Mustard

(2) *Datura*

(3) Chilli

(4) Pea

15) Read the following statements carefully.
A. Leaves originate from shoot apical meristem.
B. Opposite type of phyllotaxy are found in chinarose.
C. Peach is a perigynous flower.
D. A fertile stamen is called staminode.
E. *Dianthus* are example of axile placentation.
F. Fruit of Solanaceae family is berry or capsule.
G. Coconut develop from monocarpellary superior ovaries and are one seeded.
Find out the correct options from above mentioned statement

(1) A, B, C, D and E

(2) A, C, F and G

(3) A, C, D and F

(4) B, D and E

16) Epicalyx is commonly found in the family :

(1) Poaceae

(2) Malvaceae

(3) Asteraceae

(4) Cucurbitaceae

17) When xylem and pholem are arranged in an alternate manner along the different radii, the arrangement is called___such as in roots.

(1) Cojoint

(2) Bicolateral

(3) Concentric

(4) Radial

18) Match the following :

| | **I** | | **II** |
|---|---|---|---|
| (i) | Meristem present at the tips of roots & shoots. | (a) | Axillary buds |

| | | | |
|---|---|---|---|
| (ii) | Cells left behind from shoot apical mesistem & present in the axils of leaves | (b) | Lateral meristem |
| (iii) | Meristem occurs between mature tissue | (c) | Apical meristem |
| (iv) | Fasicular vascular cambium | (d) | Intercalary meristem |

(1) i-c, ii-a, iii-b, iv-d

(2) i-d, ii-a, iii-b, iv-c

(3) i-c, ii-a, iii-d, iv-b

(4) i-a, ii-d, iii-b, iv-c

19) Which one is not an example of lateral meristem ?

(1) Fasicular vascular cambium

(2) Intercalary meristem

(3) Interfasicular cambium

(4) Cork-cambium

20) Match the following -

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (a) | Tracheids | (i) | Highly thickened walled structure with obliterated central lumen |
| (b) | Vessels | (ii) | Living and thin walled |
| (c) | Xylem fibres | (iii) | Elongated or tube like cells with thick and lignified wall |
| (d) | Xylem parenchyma | (iv) | Long cylindrical tube like structure with perforation plate |

(1) (a–i), (b–ii), (c–iii), (d–iv)

(2) (a–iii), (b–iv), (c–i), (d–ii)

(3) (a–ii), (b–iii), (c–i), (d–iv)

(4) (a–iv), (b–iii), (c–ii), (d–i)

21) The _____ help in maintaining the pressure gradient in the sieve tubes.

(1) guard cells

(2) subsidiary cells

(3) companion cells

(4) albuminous cells

22) **Statement-I :** The collenchyma occur below the epidermis in the stems of dicotyledonous plants.
**Statement-II :** Sclerenchyma provide mechanical support to the growing parts of the plant such as young stem and petiole of a leaf.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct
(4) Both **Statement I** and **Statement II** are correct.

23) How many in the list given below are the part of ground tissue system ?
Epidermis, Root hairs, Xylem, Phloem, Cortex, Pericycle, Pith.

(1) One
(2) Two
(3) Three
(4) Four

24) In dorsiventral leaves, the location of xylem and spongy tissue are towards :-

(1) Abaxial epidermis
(2) Adaxial Epidermis
(3) Adaxial and abaxial side respectively
(4) Abaxial and adaxial side respectively

25) **Assertion (A) :** Moncotyledonous roots do not undergo any secondary growth.
**Reason (R) :** Vascular bundles are conjoint and closed in monocotyledonous root.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

26) A dicot root differs from a monocot root in which of the following ?

(1) Presence of hypodermis
(2) Presence of epidermis
(3) Presence of ill-developed (poorly developed) pith
(4) Presence of radial vascular bundle

27) Casparian strip are found in :

(1) Epidermis
(2) Endodermis
(3) Endothecium

(4) Pericycle

28) (i) The phloem parenchyma present in monocotyledonous stem.
(ii) Phloem fibres are made up of collenchyma tissue.
(iii) The stomatal aperture, guard cells and subsidiary cell are together called stomatal appratus.
(iv) Trichomes are usually multicellular but arise from the hypodermis parts.
(v) Root hairs play important role in absorbing water and minerals from the soil.
How many of the following statements are incorrect ?

(1) 5
(2) 4
(3) 3
(4) 2

29) Embryosac is :-

(1) Megasporangium
(2) Megaspore
(3) Female gametophyte
(4) Female gamete

30) The cell which divide by mitosis to form embryosac is ?

(1) Megaspore mother cell
(2) Microspore mother cell
(3) Functional megaspore
(4) Nucellus

31) In angiosperm triple fusion is necessary for the formation of ?

(1) Seed coat
(2) Fruit wall
(3) Embryo
(4) Endosperm

32) In most of the angiosperms, pollen grains are shed off at -

(1) 2-celled & 2-nucleate stage
(2) 2-celled & 3-nucleate stage
(3) 7-celled & 8-nucleate stage
(4) 3-celled & 3-nucleate stage

33) Which of the following plant/plants produce both chasmogamous and cleistogamous flowers ?

(1) *Viola*
(2) *Oxalis*
(3) *Commelina*

(4) All of these

34) Select incorrect statement regarding pollination.

(1) Animals are attracted to flower by colour or fragrance.
(2) Nectar and pollen grains are the usual floral rewards
(3) Insect pollinated flowers are mostly small and rich in nectar and fragrance.
(4) Pollen grains are protected from wetting by mucilaginous covering.

35) A flower secrete foul odours pollinated by :

(1) Birds
(2) Flies and beetle
(3) Bat
(4) Elephant

36) In angiosperm the diploid structure of fertilized ovules are :

(1) Antipodal cell, synergid, poller nuclei
(2) Integument, nucellus, endosperm
(3) Zygote, integument, nucellus
(4) Egg cell, Antipodal cell, synergid

37) (i) Stamen is equivalent to microsporophyll
(ii) Whorl of stamens is called androecium.
(iii) The proximal end of filament is attached to the thalamus or petals.
(iv) A typical angiosperm anther is bilobed with each lobe having four theca i.e they are dithecous.
(v) A typical anther is bilobed and tetrasporangiate
How many of the above statement is/are incorrect ?

(1) Four
(2) Two
(3) Three
(4) One

38) If both male and female flowers are present on same plant but flower are unisexual then

(1) It prevent autogamy
(2) It prevent geitonogamy
(3) It prevent xenogamy
(4) It prevent both autogamy and geitonogamy

39) How many meiosis and mitotic division are respectively required for the formation of mature male gametophyte from microspore mother cell in angiosperm ?

(1) 1 and 1
(2) 1 and 2

(3) 2 and 1

(4) 2 and 2

40) A flower of Mustard has six stamens and if each microsporangium has 40 pollen mother cells, then how many pollen grains will be formed in the flower ?

(1) 2400

(2) 1920

(3) 3840

(4) 960

41) How many meiotic divisions are required for the formation of 200 grains of maize?

(1) 200

(2) 400

(3) 250

(4) 300

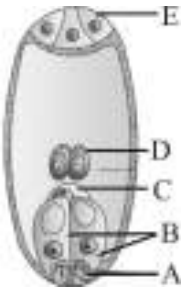
42) Apomictic embryos in *Citrus* and *Mangifera* develops from :-

(1) Diploid egg

(2) Haploid egg

(3) Maternal sporophytic tissue of ovule

(4) Maternal gametophytic tissue of ovule

43) 

In above diagram which of the following is correct function of the corresponding labelling :-

(1) C - Formation of primary endosperm.

(2) E - Entry of pollen tube in embryosac.

(3) A - Guide the entry of pollen tube.

(4) D - Formation of zygote after fusing with one of the male gamete.

44) (i) Scutellum (ii) Radicle
(iii) Coleoptile (iv) Plumule
How many structure from above list are diploid ?

(1) Three

(2) Four

(3) Two

(4) One

45) How many chromosomes are present in egg, embryo and endosperm respectively of hexaploid wheat (2n = 6x = 42) plant ?

(1) 21, 42, 63
(2) 7, 14, 21
(3) 21, 42, 21
(4) 42, 42, 63

46) Supporting cells found in the germinal epithelium of testis are called as :-

(1) Leydig cells
(2) Sertoli cells
(3) Granular cells
(4) Phagocytic cells

47) Which one of the following is the correct matching of the events occuring during menstrual cycle ?

(1) Menstruation : Breakdown of myometrium and ovum not fertilised
(2) Ovulation : LH and FSH attain peak level and sharp fall in the secretion of progesterone.
(3) Proliferative phase : Rapid regeneration of myometrium and maturation of Graafian follicle.
(4) Development of corpus luteum : Secretory phase and increased secretion of progesterone.

48) Read Column-I & Column-II and identify their correct match-

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| A. | Gametogenesis | i. | Transport of sperms into female genital tract |
| B. | Insemination | ii. | Formation of gametes |
| C. | Fertilisation | iii. | Blastocyst attachment to uterine wall |
| D. | Gestation | iv. | Fusion of male & female gamete |
| | | v. | Embryonic development |
| | | vi. | Delivery of baby |
| | | vii. | Release of semen from penis |

(1) (A–iv), (B–i), (C–ii), (D–vi)

(2) (A–ii), (B–i), (C–iv), (D–v)

(3) (A–ii), (B–vii), (C–i), (D–vi)

(4) (A–ii), (B–vii), (C–iv), (D–v)

49) Given below are two statements :
**Assertion (A) :** Breast-feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby.
**Reason (R) :** The yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant.
In the light of the above statements, choose the correct answer from the options given below :-

(1) Both (A) and (R) are correct but (R) is not the correct explanation (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct but (R) is the correct explanation (A).

50) Match list-I with list-II relating to human female external genitalia :-

| | **List-I (Structures)** | | **List-II (Features)** |
|---|---|---|---|
| A. | Mons pubis | I. | Fleshy folds of tissue which surround the vaginal opening |
| B. | Hymen | II. | Cushion of fatty tissue covered by skin and hair |
| C. | Clitoris | III. | Tiny finger-like structure above the urethral opening |
| D. | Labia majora | IV. | Membrane which is often torn during first coitus |

Choose the correct answer from the options given below :-

(1) A-II, B-III, C-IV, D-I

(2) A-IV, B-III, C-II, D-I

(3) A-I, B-IV, C-II, D-II

(4) A-II, B-IV, C-III, D-I

51) Which of the following structure is not include in female external genitalia ?

(1) Clitoris

(2) Vagina

(3) Mons pubis

(4) Hymen

52) The.....is a tiny finger like structure which lies at the upper junction of the two labia minora above the urethral opening.

(1) Mons pubis

(2) Labia majora

(3) Clitoris

(4) Hymen

53) **Assertion :-** Endometrium is necessary for implantation of blastocyst.
**Reason :-** In the absence of fertilization, the corpus luteum degenerates that causes disintegration of myometrium.

(1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(2) Assertion is True but the Reason is False.

(3) Assertion is False but the Reason is True.

(4) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

54) **Statement-I :** The presence or absence of hymen is not a reliable indicator of virginity.
**Statement-II :** The hymen is often torn during the first coitus.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

55)

Match the columns and select the correct option :-

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | Head | (I) | Hydrolytic enzymes |
| (B) | Middle piece | (II) | Sperm motility |
| (C) | Acrosome | (III) | Nebenkern |
| (D) | Tail | (IV) | Genetic material |

(1) A-ii, B-iv, C-i, D-iii

(2) A-iv, B-iii, C-i, D-ii

(3) A-iv, B-i, C-ii, D-iii

(4) A-ii, B-i, C-iii, D-iv

56) The glandular tissue of each breast is divided into .....A..... , .....B..... containing clusters of cells called alveoli.

(1) A-15-20, B-mammary duct

(2) A-20-25, B-mammary lobes

(3) A-15-20, B-mammary lobes

(4) A-5-10, B-mammary tubules

57) Which of the following statement is NOT correct?

(1) Gametogenesis begins in the primary sex organ at the time of puberty in male and female.
(2) Each spermatogonium is diploid and contains 46 chromosomes.
(3) After spermiogenesis sperm heads become embedded in the sertoli cells.
(4) At puberty only 60000-80000 primary follicles are left in each ovary.

58) Oogenesis is initiated during the _____ when a couple of million gamete mother cells are formed within each ovary.

(1) Embyonic development
(2) Puberty
(3) Old age
(4) Infant stage

59) How many spermatozoa are produced from 150 secondary spermatocyte ?

(1) 600
(2) 150
(3) 300
(4) 75

60) The mammary gland of the female undergo differentiation during pregnancy and starts producing milk towards the end of pregnancy by the process called ?

(1) Lactation
(2) Colostrum
(3) Parturation
(4) Menarche

61) Which of the following statement is incorrect ?

(1) Secretions of bulbourethral glands helps in lubrication of penis during coitus.
(2) Release of sperm form seminiferous tubules is known as spermiation..
(3) Inner cell mass present in blastocyst gets differentiated as embryo.
(4) By end of 12 weeks first movement of foetes is observed.

62) **Assertion :-** Cells of Alveoli of breast secretes milk which is stored in mammary tubules.
**Reason :-** Mammary gland are paired structure that contains glandular tissues and fats.

(1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(2) Assertion is False but the Reason is True.
(3) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(4) Assertion is True but the Reason is False.

63) **Assertion :-** During fertilization, a sperm comes in contact with zona pellucida layer of the ovum and induces changes in membrane that block entry of additional sperms.
**Reason :-** Sex of the baby is determined by the father and not by the mother.

(1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(2) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Assertion is False but the Reason is True.

64) Which of the following hormone is not increased during parturition?

(1) Estrogen
(2) Cortisol
(3) Progesterone
(4) Oxytocin

65) Which part of oviduct joins the uterus ?

(1) Fimbrae
(2) Infundibulum
(3) Ampulla
(4) Isthmus

66) **Assertion (A) :** During pregnancy all events of the menstrual cycle stop and there is no menstruation.
**Reason (R) :** The corpus luteum secretes large amount of progesterone which is essential for maintenance of the endometrium.

(1) A is correct but R is not correct.
(2) A is not correct but R is correct.
(3) Both A and R are correct and R is the correct explanation of A.
(4) Both A and R are correct and R is not the correct explanation of A.

67) Match the list-I with list-II and choose the correct match :-

| | **List-I** | | **List-II** |
|---|---|---|---|
| (A) | High level of LH | (I) | Follicular development |
| (B) | Low level of LH | (II) | Release the ovum |
| (C) | High level of FSH | (III) | Corpus luteum degenerate |
| (D) | High level of progesterone | (IV) | Maintain endometrium |

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-III, C-I, D-IV

(3) A-II, B-III, C-IV, D-I

(4) A-I, B-II, C-III, D-IV

68) **Statement-I :** Each spermetogonia is diploid cell and contain 46 chromosomes.
**Statement-II :** LH acts on sertoli cells and stimulate synthesis & secretion of estrogen.

(1) Statement-I is correct but statement-II is incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Both statement-I and statement-II are correct

(4) Both statement-I and statement-II are incorrect

69) **Assertion :-** During the mid of menstrual cycle, rupture of graafian follicle and thereby the release of ovum occurs.
**Reason :-** Rapid secretion of LH leading to its maximum level during the mid-menstrual cycle called LH surge.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Assertion is False but the Reason is True.

70) Which of the following hormone is responsible for degeneration of corpus luteum?

(1) High level of estrogen

(2) Low level of progesterone

(3) High level of LH

(4) Low level of LH

71) Find the odd one out with respect to hormone produced in women only during pregnancy.

(1) hCG

(2) Estrogen

(3) hPL

(4) Relaxin

72) **Statement-I :** .The inner cell mass contain certain cells called stem cells which have the potency to give rise to all the tissue and organs.
**Statement-II :** The inner cell mass get differentiated as the embryo.

(1) Both statement-I and statement-II are incorrect.

(2) Both statement-I and statement-II are correct.

(3) Only statement-I is correct

(4) Only statement-II is incorrect

73)

Secretions from which of the following structure is essential for maturation & motility of sperms ?

(1) Epididymis
(2) Vas deferens
(3) Prostate
(4) All of these

74) What would happen in the absence of fertilization ?

(1) FSH and LH will be high
(2) Corpus luteum degenerates
(3) Progesterone levels will remain high
(4) Corpus luteum persist

75) Match list I with list II :

| | **List I** | | **List II** |
|---|---|---|---|
| (A) | Lubrication of penis | (i) | Seminal vesicles |
| (B) | Collection of ovum | (ii) | Bulbourethral gland |
| (C) | Nutrition to germ cells | (iii) | Leydig cells |
| (D) | Differentiated as embryo | (iv) | Sertoli cells |
| | | (v) | Inner cell mass |
| | | (vi) | Fimbriae |
| | | (vii) | Ampulla |

Choose correct answer from option given below :
(1) A-i, B-vii, C-iv, D-v
(2) A-iii, B-v, C-ii, D-vii
(3) A-ii, B-vi, C-iv, D-v
(4) A-ii, B-vii, C-i, D-iii

76)

In India, family planning programmes were initiated in -

(1) 1951
(2) 1971
(3) 2017
(4) 1963

77) Match Column-I with Column-II :

| | Column-I | | Column-II |
|---|---|---|---|
| A. | Under the skin | (i) | Non-steroid |
| B. | Saheli | (ii) | LNG-20 |
| C. | Uterus | (iii) | Implant |
| D. | Cover the cervix | (iv) | Diaphragm |

(1) (A–iii), (B–i), (C–iv), (D–ii)

(2) (A–i), (B–iii), (C–ii), (D–iv)

(3) (A–iii), (B–i), (C–ii), (D–iv)

(4) (A–iv), (B–ii), (C–iii), (D–i)

78) Read the following terminology of column-I & II and identify their correct match :-

| | Column-I | | Column-II |
|---|---|---|---|
| (A) | Lactational amenorrhea | (i) | Barrier method |
| (B) | Cervical cap | (ii) | Copper releasing IUD |
| (C) | Lippes loop | (iii) | Natural method |
| (D) | Multiload 375 | (iv) | Non-medicated IUD |

(1) A-iii, B-i, C-iii, D-iv

(2) A-i, B-iii, C-ii, D-iv

(3) A-i, B-ii, C-iv, D-iii

(4) A-iii, B-i, C-iv, D-ii

79)

For embryo transfer, embryo is obtained from

(1) IVF

(2) ICSI

(3) In vivo fertilization

(4) All of these

80) **Statement-I :** Barrier method is a natural contraceptive method which works on the principle of avoiding chances of ovum and sperm meeting.
**Statement-II :** Administration of progestogens or progestogen-estrogen combinations or IUDs within 72 hours of coitus have been found to be very effective as emergency contraceptives.

(1) Statement-I is correct and statement-II is incorrect.

(2) Both statement-I and statement-II are correct.

(3) Statement-I is incorrect and statement-II is correct.

(4) Both statement-I and statement-II are incorrect.

81) The medical termination of pregnancy (amendment) act, 2017 was enacted by the government of India with the intention of :

(1) Reducing the incidence of illegal abortion and consequent maternal mortality and morbidity

(2) Increasing number of couples with small families

(3) Better detection and cure of STD

(4) Increasing number of medically assisted deliveries and better post natal care leading to decreased maternal and infant mortality rates.

82) Given below are two statements.
**Statement-I :-** Lactational amenorrhea has been reported to be effective only upto a maximum period of six months following parturition.
**Statement-II :-** Implants only works on principle of prevention of gametes (sperms & ovum) from physically meeting with help of barriers.
Choose correct answer from option given below :-

(1) Both statement-I and statement-II are incorrect.

(2) Statement-I is incorrect but statement-II is correct.

(3) Both statement-I and statement-II are correct.

(4) Statement-I is correct but statement-II is incorrect.

83) Read the following statements :-
(a) Combination pills have to be taken daily for a period of 21 days starting preferably within the first five days of menstrual cycle.
(b) Pills inhibit ovulation and implantation as well as alter the quality of cervical mucus to prevent/retard entry of sperms.
(c) Saheli-the new oral contraceptive for the females contains a non-steroidal preparation
(d) Withdrawal or coitus interruptus is method in which the male partner withdraws his penis from the vagina just after ejaculation so as to avoid insemination.
(e) IUDs are inserted by doctors or expert nurses in the fallopian tube through vagina.
Choose the correct statements :-

(1) a, b & c

(2) b, c & d

(3) a, c & d

(4) a, b & d

84)

In vasectomy a small part of vas deferens is removed & tied up through a small incision on :

(1) Scrotum

(2) Abdomen

(3) Thorax

(4) Either 1 or 3

85) Which of the following statements are correct ?
(a) Amniocentesis is used to test for presence of certain disorders like down's syndrome.

(b) Saheli contains small dose of either progesteron or progesteron-estrogen combination.
(c) Infections like hepatitis-B and HIV can also be transmitted through surgical instruments.
(d) Medical termination of pregnancy cannot be done on failure of contraceptive during coitus.
(e) In ZIFT, embryo's with more than 8 blastomeres are transferred in fallopian tube.

(1) a and c only
(2) b, c and d
(3) a, c and e
(4) c, d and e

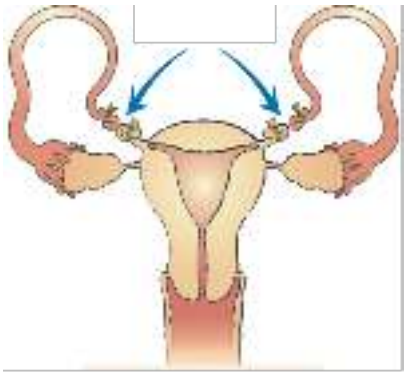
86) What is false for GIFT ?

(1) It is called gamete intra fallopian transfer.
(2) Ovum from a donor is transferred into oviduct of the recipient
(3) Zygote from a donor is transferred into oviduct of the recipient
(4) The recipient may not be able to produce ovum.

87) **Statement-I :** Intracytoplasmic sperm injection (ICSI) is another specialized procedure of in vivo fertilization.
**Statement-II :** Infertility cases due to inability of the male partner to inseminate the female can be corrected by artificial insemination.

(1) Statement-I is correct
(2) Statement-II is correct
(3) Both statement-I and statement-II are correct
(4) Both statement-I and statement-II are incorrect

88) Identify the correct statement for given diagram :-



(1) Fallopian tubes are cut and removed
(2) Vas deferens is cut and tied
(3) Oviducts are cut and tied
(4) It is a type of barrier contraceptive method.

89) Statutory ban on amniocentesis helps to prevent :-

(1) Female foeticides
(2) Sexually transmitted disease
(3) Genetic disorders in foetus

(4) All of these

90) Given below are two statements.
**Assertion :-** Condoms and vaults are barrier methods of contraception.
**Reason :-** They prevent physical meeting of sperms and ova.

(1) Both A and R are true and R is NOT the correct explanation of A.

(2) A is true and R is false.

(3) A is false and R is true.

(4) Both A and R are true and R is the correct explanation of A.

ANSWER KEYS

## PHYSICS

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 3 | 1 | 1 | 3 | 1 | 4 | 3 | 1 | 3 | 3 | 3 | 3 | 2 | 2 | 1 | 1 | 1 | 2 | 3 | 4 |
| Q. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| A. | 2 | 1 | 4 | 4 | 3 | 4 | 1 | 1 | 4 | 4 | 4 | 2 | 2 | 1 | 3 | 1 | 2 | 1 | 3 | 4 |
| Q. | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| A. | 1 | 3 | 1 | 3 | 2 | | | | | | | | | | | | | | | |

## CHEMISTRY

| Q. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 3 | 1 | 4 | 1 | 3 | 1 | 1 | 3 | 2 | 4 | 3 | 2 | 2 | 4 | 1 | 4 | 2 | 1 | 3 | 3 |
| Q. | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |
| A. | 1 | 2 | 4 | 1 | 3 | 1 | 4 | 1 | 2 | 4 | 3 | 4 | 3 | 2 | 2 | 4 | 3 | 2 | 4 | 1 |
| Q. | 86 | 87 | 88 | 89 | 90 | | | | | | | | | | | | | | | |
| A. | 4 | 3 | 3 | 3 | 2 | | | | | | | | | | | | | | | |

## BIOLOGY

| Q. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 1 | 3 | 1 | 4 | 2 | 4 | 2 | 3 | 4 | 2 | 4 | 2 | 1 | 4 | 2 | 2 | 4 | 3 | 2 | 2 |
| Q. | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 |
| A. | 3 | 2 | 3 | 3 | 2 | 3 | 2 | 3 | 3 | 3 | 4 | 1 | 4 | 3 | 2 | 3 | 4 | 1 | 2 | 3 |
| Q. | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| A. | 3 | 3 | 3 | 2 | 1 | 2 | 4 | 2 | 4 | 4 | 2 | 3 | 2 | 1 | 2 | 3 | 1 | 1 | 3 | 1 |
| Q. | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 |
| A. | 4 | 2 | 1 | 3 | 4 | 3 | 2 | 1 | 1 | 4 | 2 | 2 | 4 | 2 | 3 | 1 | 3 | 4 | 4 | 3 |
| Q. | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | | | | | | | | | | |
| A. | 1 | 4 | 1 | 1 | 1 | 3 | 2 | 3 | 1 | 4 | | | | | | | | | | |

SOLUTIONS

PHYSICS

1)

Initial force, $F = \frac{KQ^2}{r^2}$ ...(1)

New force, $4F = \frac{K(Q+2)(Q)}{r^2}$ ...(2)

From (1) and (2)

$(Q + 2)(Q) = 4Q^2$

$\Rightarrow Q + 2 = 4Q$

$2 = 3Q$

$\Rightarrow \frac{2}{3}C$

2) (A) $E_{ring} = \frac{kQr}{(R^2 + r^2)^{3/2}}$

$E_{(r=0)} = 0$, zero at centre

E increases to maximum at a distance of $r = \frac{R}{\sqrt{2}}$ and then for from ring, at behaves like due to a point charge

$E_{(r>>R)} \approx \frac{kQ}{x^2} \propto \frac{1}{x^2}$

Graph (Q) represents this behaviour.

(A)-(Q)

(B) For a uniformly charged sphere

$E_{(r<R)} = \frac{kQr}{R^3} \propto r$

$E_{(r=R)} = \frac{kQ}{R^2}$(max), $E_{(r>R)} = \frac{KQ}{r^2} \propto \frac{1}{r^2}$

Graph (S) represents this behaviour.

(B)-(S)

(C) For a uniformly charged spherical shell:

$E_{(r<R)} = 0$; $E_{(r=R)} = \frac{kQ}{R^2} \Rightarrow$ (Max)

$E_{(r>R)} = \frac{kQ}{r^2} \propto \frac{1}{r^2}$

Graph (R) represents this behaviour.

(D) Combination of equal and opposite charge +Q and –Q may be treated as dipole (but distance may not be assumed very small). At centre, the electric field is maximum and then decreases with distance r from bisector as

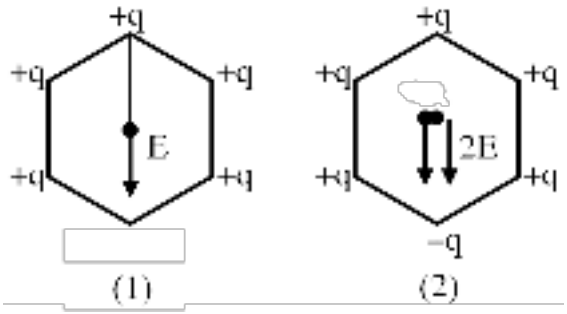
$E_{eq.} = \frac{kPr}{(R^2 + r^2)^{3/2}}$

Graph (P) represents this behaviours.

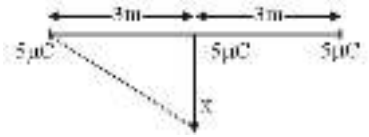
3) $\tan\theta = \frac{F_E}{mg} = \frac{\sigma q}{2\varepsilon_0 mg} \Rightarrow \theta = \tan^{-1}\left(\frac{\sigma q}{2\varepsilon_0 mg}\right)$

4) All combinations are based on following properties of electric field lines : (a) Tangent to electric field line represent the direction of resultant electric field at a point.
(b) Field lines originate from a positive charge and terminate on negative charge.
(c) Number of field lines associated with a charge is directly proportional to the magnitude of charge.
(d) Density of field lines is directly proportional to the strength of electric field.

5) $E_1 = \frac{kq}{r^2}$ $E_2 = \frac{2kq}{r^2} = 2E_1$




6)




$$\frac{-k\times 5\times 5\times 10^{-12}}{|3|}\times 2 + 0.06 = -\frac{k\times 5\times 5\times 10^{-12}\times 2}{\sqrt{(3^2+x^2)}}$$

$$\Rightarrow -\frac{9\times 10^{-9}\times 25\times 10^{-12}\times 2}{3} + 0.06 = \frac{-9\times 10^{9}\times 25\times 10^{-12}\times 2}{\sqrt{3^2+x^2}}$$

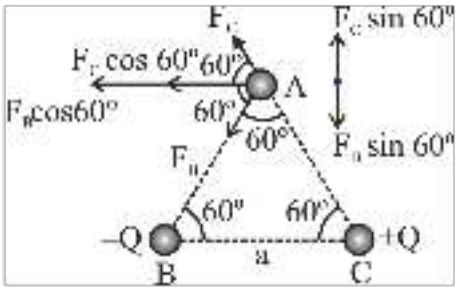
$$\Rightarrow -0.15 + 0.06 = -\frac{0.45}{\sqrt{3^2+x^2}}$$

$$-0.09 = -\frac{0.45}{\sqrt{3^2+x^2}}$$

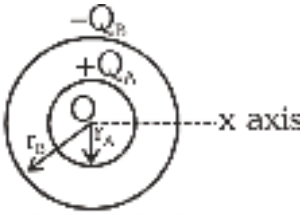
$x = 4$

7) $\vec{F_B} = \vec{F_C} = k.\frac{Q^2}{a^2}$

$\mathbf{F}_{net}$ =




Hence force experienced by the charge at A in the direction normal to BC is zero.




8) $|Q_B| > |Q_A|$

E = 0 ; $r < r_A$

$E = \frac{kQ_A}{r^2}$ ; $r_A < r < r_B$ (along + x)

$E = \frac{-k(Q_B - Q_A)}{r^2}$ ; $r > r_B$ (along – x)

9)

Due to transfer of electron weight may increase or decrease.

10) By connecting wire, all charge flow on outside sphere (A)

So $\sigma_A = \frac{Q_1 + Q_2}{4\pi r_1^2}$ $\sigma_B = 0$

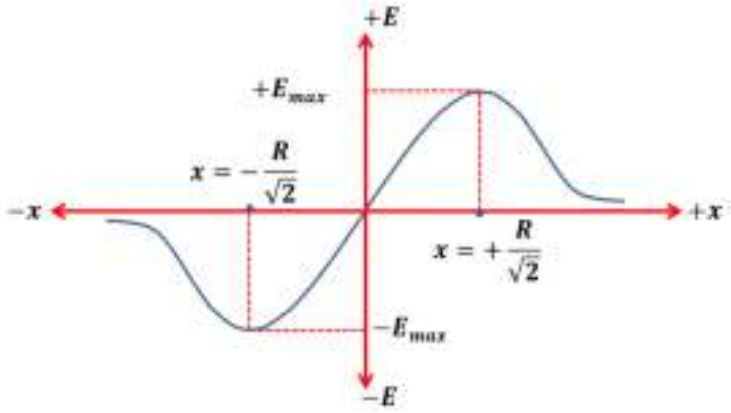
11) X-ray ionize the gas, so become partially conducting. The charge on leaves will start leaking and they will collapse gradually.

12)

For cube $\phi_1 = \frac{a\sqrt{3}\lambda}{\varepsilon_0}$, sphere $\phi_2 = \frac{2a\lambda}{\varepsilon_0}$

$\frac{\phi_1}{\phi_2} = \frac{\sqrt{3}}{2}$

13)




14)

$E_x = -\frac{dV}{dx} E_y = -\frac{dV}{dy}$ where $E_x, E_y$ are the electric field components, and $\frac{dV}{dx}, \frac{dV}{dy}$ are the rate of change of potential with respect to x and y.

-Potential difference $\Delta V_x = 3\,V$ - Distance $\Delta x = 2.0 \times 10^{-2} m$

$$\frac{dV}{dx} = \frac{3}{2.0 \times 10^{-2}} = 150\,V/m$$

Then, $E_x = -150\,V/m$

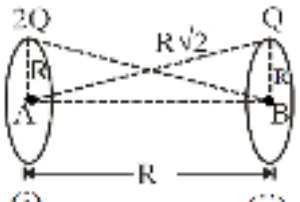
- Potential difference $\Delta V_y = -3\,V$ (negative sign matters) - Distance $\Delta y = 1.0 \times 10^{-2} m$

$$\frac{dV}{dy} = \frac{-3}{1.0 \times 10^{-2}} = -300\,V/m$$

Then, $E_y = -(-300) = +300\,V/m$

So the components of the electric field are:

$E_x = -150\,V/m, \quad E_y = +300\,V/m$




15) **Explain Question:** We have to calculate potential at point 'A'.

**Concept:**

Potential at a point due to circular ring $= \frac{kQ}{d}$

**Solution:**

For ring (i) $\Rightarrow$ d =R

for ring (ii) $\Rightarrow d = \sqrt{2}R$

$V_{total} = V_{2Q} + V_{-Q}$

$$V_{total} = \frac{k(2Q)}{R} + \frac{k(-Q)}{\sqrt{2}R}$$

$$V_{total} = \frac{kQ}{R}\left[2 - \frac{1}{\sqrt{2}}\right]$$

$$V_{total} = \frac{kQ}{R} \times \frac{\left[2\sqrt{2} - 1\right]}{\sqrt{2}}$$

$$V_{total} = \frac{Q\left(2\sqrt{2} - 1\right)}{4\pi\varepsilon_0 \sqrt{2}R}$$

**Final Answer:** Option (1)

16) $\vec{\tau} = \vec{P} \times \vec{E}$

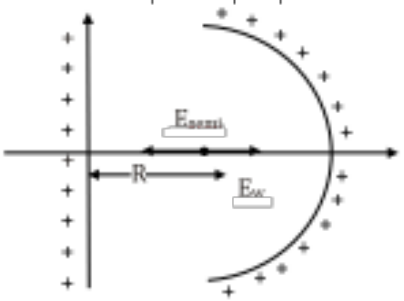
$$= \left(\hat{i} + 2\hat{j}\right) \times \left(\hat{i} + 2\hat{j} + 3\hat{k}\right)$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 0 \\ 1 & 2 & 3 \end{vmatrix}$$

17) Electric field intensity at a point is the vector sum of all individual EF at a point.

$\therefore \vec{E} = \vec{E}_{\text{due to infinite wire}} + \vec{E}_{\text{due to semicirculation}}$

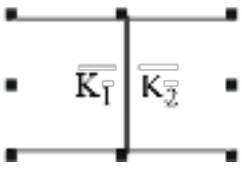
$\because \vec{E} = 0 \Rightarrow \left|\vec{E}_{wire}\right| = \left|\vec{E}_{semicircular}\right|$




$$\frac{2k\lambda_1}{R} = \frac{2k\lambda_2}{R}\sin\left(\frac{\theta}{2}\right)$$

$$\frac{\lambda_1}{\lambda_1} = \sin\left(\frac{\theta}{2}\right)$$

Here $\theta = \pi \Rightarrow$ $\therefore \boxed{\frac{\lambda_1}{\lambda_1} = 1}$




18)

$$K_1 = \frac{2 \times 1 \times 3}{1 + 3} = \frac{3}{2}$$

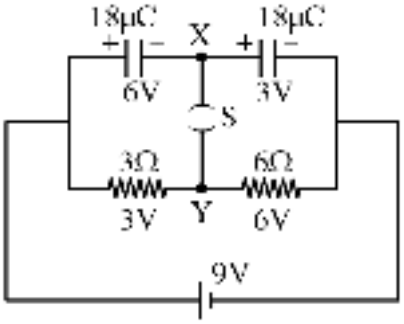
$$K_2 = \frac{2 \times 2 \times 4}{2 + 4} = \frac{8}{3}$$

$$K_{eq} = \frac{K_1 + K_2}{2} = \frac{\frac{3}{2} + \frac{8}{3}}{2} = \frac{9 + 16}{12} = \frac{25}{12}$$

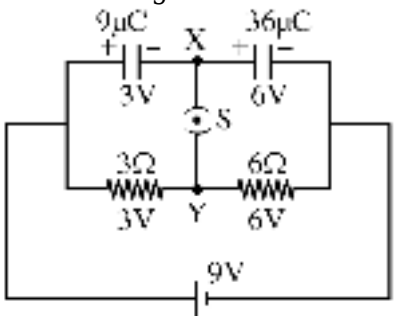
$$C = \frac{\varepsilon_0 K_{eq} A}{2d} = \frac{\varepsilon_0 A}{2d} \times \frac{25}{12} = \frac{25}{24}\frac{\varepsilon_0 A}{d}$$

19)

Before closing switch's




Net charge at function
X = +18μC – 18μC = 0C
After closing switch S



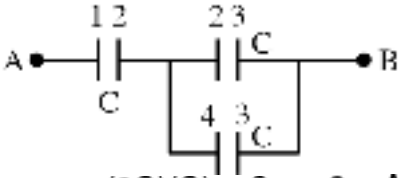

Net charge at junction
X = +36 μC – 9μC
= +27μC
Charge that flows from Y to X
= (+27 μC – 0) = +27 μC

20)

$$V_{MN} = \frac{C}{C + C} \times 60 = 30V$$

21)

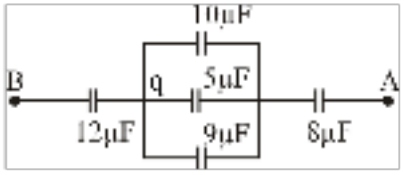
Equivalent circuit can be redrawn as :




$$\Rightarrow C_{eq} = \frac{(2C)(C)}{2C + C} = \frac{2}{3}C = \frac{2}{3}\frac{\varepsilon_0 A}{d}$$

$$\Rightarrow \theta = C_{eq}E = \frac{2}{3}\frac{\varepsilon_0 AE}{d}$$

22) Battery is removed, charge on plate is constant. By placing a dielectric slab between the plates, capacitance increase so the potential difference between the plates and energy stored will decreases.

23) 


$$\frac{1}{C_{eq}} = \frac{1}{12} + \frac{1}{24} + \frac{1}{8}$$

Now

$C_{eq} = 4\mu F$

for charge the potential across 5μF is 10 volts so q = 50μC

24)

Electric field between plates

$$E = \frac{\sigma}{K\varepsilon_0} \Rightarrow \sigma = E(K\varepsilon^0)$$

$\sigma = (3 \times 10^4)(2.2 \times 8.85 \times 10^{-12})$ C/m$^2$

$= 58.4 \times 10^{-8} \approx 6 \times 10^{-7}$ C/m$^2$

25) $C \propto \frac{ab}{b-a}$; a = R – x, b = R so, $C \propto \frac{R(R-x)}{x}$

26) $i_g = 100 \times 10^{-4} = 0.01$ A

$V = I_g(R + R_g)$

$10 = 0.01 ( R + 100)$

$\Rightarrow R = 900\ \Omega$

27) $V_C + 5 + 12 – 12 – 3 – 8 = V_B$

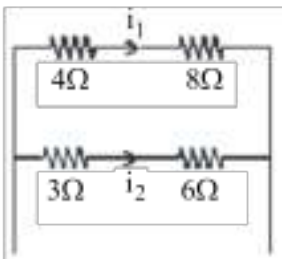
$V_C – V_B = 6V$

28) In series current will be same so

$$\frac{P_1}{P_2} = \frac{I^2R_1}{I^2R_2} = \frac{R_1}{R_2} = \frac{1}{2}$$

29) If current in P and Q equal to I then current in R and S will be 2I.

H = (Current$^2$ × resistance) which is greatest for S.

30) 

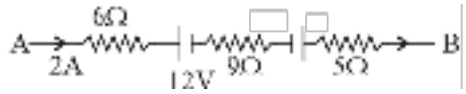

$$\frac{4}{8} = \frac{3}{6}$$

So this is balance wheat stone bridge

 No current in 5Ω

$I \propto \frac{1}{R}$ so

$\therefore \frac{i_1}{i_2} = \frac{9}{12} = \frac{3}{4}$

31) 


This is a series connection. Further, whatever current enters A has to pass B. I = 2A.
This total resistance = 6 + 9 + 5 = 20Ω. The effective potential across the resistances is 20Ω × 2A = 40V. But (+12 – 4) V is opposing the potential difference across AB therefore the potential difference applied across AB is 40V + 8V = 48V. **OR** You can apply KVL

32)

In case of discharging of a cell
V = E – ir

For i = 0 ; V = E and For $i = \frac{E}{r}$: V = 0

33) $\frac{R_1 + 10}{R_2} = \frac{50}{100-50} = 1$

$\Rightarrow R_2 = R_1 + 10$ ...(1)

$\Rightarrow$ Now without 10 Ω, $\frac{R_1}{R_2} = \frac{40}{60}$

$\Rightarrow R_2 = \frac{3}{2}R_1$ ...(2) by (1) & (2)

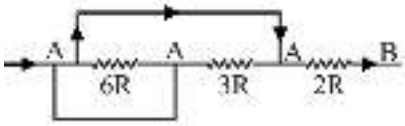
$R_1 = 20\ \Omega$

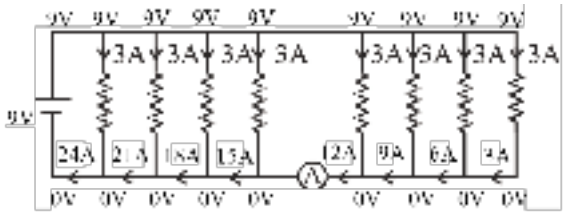
34) Temperature coefficient of resistance for manganin and constantan is very small, so their resistance approximately remain constant with change in temperature.

35) $J = \sigma E$

So $E \propto J \propto \frac{1}{A}$

36) In absence of applied potential, electrons move in random zig-zag motion, due to thermal energy with speed of order of $10^5$ m/s in all possible directions. Thus average velocity of electron is zero. Hence it does not produce electric current. Also electrons are not drifted in absence of field, as there is no pull on electrons in some definite direction. Ions are part of lattice structure and are immobile for a metallic wire (solid). In light of above, only option (1) is correct statement.

37) 


38) 


39)

$V_1$ and $V_2$ are series so same current flows through them, and they are identical so have same resistance also hence $V_1 = V_2$ but $V_3$ is sum of $V_1$ and $V_2$
so $(V_1 = V_2) < V_3$

40)

Potential difference across 20Ω ⇒ V = IR
= 0.3 × 20
= 6 volt

Total current ⇒ $\frac{6}{R_1} + \frac{6}{20} + \frac{6}{15} = 0.8\ A$
$R_1$ = 60Ω

41) Potential at centre of uniformly charge solid sphere is $V_C = \frac{\rho R^2}{2\epsilon_0}$ and potential at surface of uniformly charge solid sphere is $V_S = \frac{\rho R^2}{3\epsilon_0}$ difference between them is $V_C - V_S = \frac{\rho R^2}{6\epsilon_0}$.

42) In series combination :
$E_{net}$ = (n – 2m)E
$E_{net}$ = [5 – 2(1)]E
$E_{net}$ = 3E
$r_{Net}$ = 5r

I case

$$\frac{10}{R}=\frac{\ell}{100-\ell} \qquad \text{... (i)}$$

II case $$\frac{10}{2R}=\frac{3/4\ell}{100-3/4\ell} \qquad \text{... (ii)}$$

Form (i) and (ii)

$$2=\frac{4}{3}\frac{100-3/4\ell}{100-\ell} \Rightarrow \quad \ell=\frac{200}{3}$$

Put the value of $\ell$ in equation .... (i)

$$\frac{10}{R}=\frac{200/3}{100-\frac{200}{3}}=2 \Rightarrow R=5\Omega$$

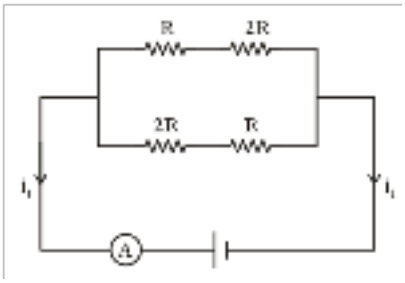
43)

44)

When key is open then

$R_{eq} = \frac{3}{2}R$

$i_1 = \frac{V}{\frac{3R}{2}} = \frac{2}{3}\left(\frac{V}{R}\right)$

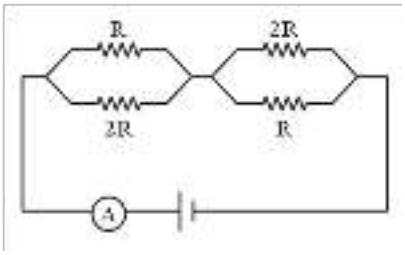



When key closed then

$R_{eq} = 2\left(\frac{2}{3}R\right) = \frac{4}{3}R$

$i_2 = \frac{V}{\frac{4}{3}R} = \frac{3}{4}\left(\frac{V}{R}\right)$

$\frac{i_1}{i_2} = \frac{2/3}{3/4} = \frac{8}{9}$




45) In balanced Wheatstone bridge, the arms of galvanometer and cell can be interchanged without affecting the balance of the bridge.

46) **Solution:**
A suitable indicator must change color within this basic pH range.
• Phenolphthalein has a pH transition range of 8.2–10, which is basic.
• Methyl orange and Methyl red have transition ranges in the acidic region.
Therefore, phenolphthalein is the correct choice.

47)

$H_2SO_4 + 2KOH \rightarrow K_2SO_4 + H_2O$
(geq) 0.04 0.03 – –
LR
0.01 1 0.03
 Heat for 1 geq = –57.2 kJ/eq
 Heat for 0.03 geq = –57.2 × 0.03
= –1.716 kJ

48)

**Solution:**

A. Cu, Ag, Au : noble metals → stable in water → suitable for arc method.
B. Na: highly reactive with water → forms explosive reaction → unsuitable for arc method.

49)

**Solution:**
$Fe^{2+} \rightarrow Fe^{3+}$; change in oxidation state = 1 → n-factor = 1;
Equivalent weight = 392 / 1 = 392

$FeSO_4.(NH_4)_2SO_4.6H_2O + KMnO_4 \xrightarrow{H^+} Mn^{+2} + Fe^{+3}$

50) $H_2SO_4$ is used. $HNO_3$ is not used because it is an auto oxidant HCl is not used because it react with $KMnO_4$.

51)

(I) Acidic medium helps keep Fe in +2 oxidation state
(II) $KMnO_4$ behave as oxidising agent in titration of Mohr's salt and convert into $Mn^{2+}$

52)

Reaction $H_2O_2 + 2I^- + 2H^+ \rightarrow 2H_2O + I_2$
Rate law $r = K[H_2O_2]\,[I^-]$

53) $n_{O_2} \times V_f = \frac{it}{96500}$

$\Rightarrow n_{O_2} = \frac{(90 \times 60)(8)}{96500 \times 4} = 0.112$

54)

**Solution:**

• In the reaction $H_2O_2$ oxidezes $I^-$ to $I_2$.

Free iodine reacts with starch to form a characteristic blue-coloured complex

This blue colour is widely used as an indicator in iodine kinetics experiments

55)

$\Rightarrow t = \frac{2.303 \times 0.6}{10^{-2}} = 138.18s$

$t = \frac{2.303}{k} \log \frac{C_o}{C_t}$

56)

Theoretical concept

57)

(A)$\log(k_2/k_1) = \frac{E_a}{2.303}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$

58)

Fact 


59)

ic = 1C, 3C, 4C

BP ⇒ Urea < $BaCl_2$ < $AlCl_3$

60)

on increasing temp solubility ↓ and $K_H$ ↑

61)

For isotonic solution osmotic pressure same.

62)

Fact

63)

$K = 0.012 \Omega^{-1} cm^{-1}$

$\frac{1}{R}\frac{\ell}{A} = 0.012$

$\frac{\ell}{A} = (0.012 \times 55)\ cm^{-1}$

$= 0.660$

64) $E^o_{cell} = E^o_{N^{n+}/N} - E^o_{M^{n+}/M}$

$= 0.81 - (-0.34) = 1.15$

65)

Formula $K_a = \frac{C\Lambda_m^2}{\Lambda_m^\infty(\Lambda_m^\infty - \Lambda_m)}$

66)

$Q = it = (0.01 \times 10^{-3})\ (10 \times 60 \times 60)$

$= 0.36$

67)

$K_{eq.} = 10^{\frac{E^o_c n}{0.06}}$

$10^{24} = 10^{\frac{E^o_c \times 6}{0.06}}$

$0.24 = E^o_C$

68)

$r = k[A]^x\ [B]^y$

$\frac{5 \times 10^{-2}}{0.2} = \left(\frac{0.3}{0.6}\right)^y$

$y = 2$

Also,

$\frac{0.2}{1.8} = \left(\frac{0.2}{0.8}\right)^x \left(\frac{0.6}{0.9}\right)^2$

$$\frac{1}{9} = \left(\frac{1}{4}\right)^{x}\left(\frac{2}{3}\right)^{2}$$
x = 1

69)

(P) Rate expression is called rate law
(Q) $r = K[A]^{o} = K$
(R) Unit of K = $Ms^{-1}$
Unit of Rate = $Ms^{-1}$
(S) Slowest step = Rate determining step

70) $K_1 = Ae^{-E_1/RT}$, $K_2 = Ae^{-E_2/RT}$
Given : $E_2 = E_1 - 2$
$$\frac{K_2}{K_1} = e^{(E_1-E_2)/RT} = e^{+\frac{2}{RT}} = e^{+\frac{1000}{3}} = 28$$
$K_2 = 28\ K_1$

71)

$$Kt = 2.303 \log\left(\frac{1}{1/3}\right)$$
$$\Rightarrow \frac{Kt}{2} = 2.303 \log\left(\frac{1}{1-x}\right)$$
$$\Rightarrow \frac{1}{1/2} = \frac{\log(3)}{\log\left(\frac{1}{1-x}\right)}$$
$$\left(\frac{1}{1-x}\right)^{2} = 3 \Rightarrow \frac{1}{1-x} = \sqrt{3}$$
$$\Rightarrow 1 - x = \text{left} = \frac{1}{\sqrt{3}}$$

72)

For isotonic solution $i_1C_1 = i_2C_2$
$$\left(3gL^{-} \times \frac{1}{Mw}\right) molL^{-1} = 2\times 5 \times 10^{-2} \text{ mol } L^{-1}$$
$$Mw = \frac{3}{10^{-1}} = 30$$

73)

$$PV = nRT = \frac{W}{Mw} \times RT$$
$$W = \frac{PV\,Mw}{RT} = \frac{1 \text{ atm} \times 0.04 \times 48}{0.0821 \times 300} = 0.078 \text{ g}$$
in 100 g water at 1 atm = 0.078 g

in 400 g at 4 atm = $\frac{0.078}{100} \times 400 \times 4$ = 1.248g

74)

Fact

75)

For dil. solution

$$\frac{P^o - P_S}{P^o} = \frac{X_B}{X_A} = \frac{m(Mw)_{solvent}}{1000}$$

$$\frac{0.5}{25} = \frac{m(18)}{1000}$$

$$\Rightarrow m = \frac{55.5 \times 0.5}{25} = 1.11$$

76)

Target reaction = reaction (2) + Reverse of reaction(1)

$$E_3^o = \frac{n_2E_2^o \times (-n_1E_1^o)}{n_3}$$

$$= \frac{2(0.34) + (1 \times -0.15)}{1}$$

= 0.53 V

77)

$$\propto = \frac{\Lambda_m}{\Lambda_m^o} = \frac{40.1}{349.6 + 54.6} = 0.099$$

78)

At equilibrium, $E_c$ = 0, n = 2 for Daniell cell

$$\Rightarrow E_c^o = \frac{0.0591}{n} \log K_c$$

$$1.1 = \frac{2.303RT}{2F} \log K_c$$

79)

$H_2$ gas and $O_2$ gas are used which are supplied externally to this cell.

80) $$r = -\frac{\Delta(A)}{\Delta t}$$

81)

$P(g) \xrightarrow{\Delta} A(g) + B(g) + C(g)$

| | | | | |
|---|---|---|---|---|
| 0.4atm | - | - | - | $t_{1/2} = 69.3$ s |
| 0.4-x | x | x | x = | $\frac{0.693}{K}$ |
| given $0.4 + 2x = P_t$ | | | K = | $\frac{0.693}{69.3}$ |
| $x = \frac{P_t - 0.4}{2}$ | | | | $= 10^{-2}$ |

$$K = \frac{1}{t}\ell n\left(\frac{P_i}{P_i - x}\right)$$

$$K = \frac{2.303}{230}\log\left(\frac{0.4}{0.4 - \frac{P_t - 0.4}{2}}\right)$$

$$10^{-2} = 10^{-2}\log\left(\frac{0.8}{1.2 - Pt}\right)$$

$$10 = \frac{0.8}{1.2 - P_t}$$

$P_t = 1.2 - 0.08 = 1.12$ atm

82)

$$ppm = \frac{w_{solute}}{w_{solution}} \times 10^6$$

$$= \frac{(25 \times 10^{-3})kg}{10^3 kg} \times 10^6 = 25\,ppm$$

83)

$d = 1.1$ g. $mL^{-1}$

20% $FeCl_3$ w/W

$$M = \frac{w}{Mw} \times \frac{1000}{V(mL)}$$

$$= \frac{20}{162} \times \frac{1000}{(100/1.1)}$$

$$= \frac{22 \times 10}{162} = 1.36$$

84)

$Rx^n$ (P) $\Rightarrow B^+$ reduced. $\Rightarrow$ SRP $C^+$ < SRP $B^+$

$Rx^n$ (Q) $\Rightarrow$ D is not oxidised $\Rightarrow$ SRP $A^+$ < SRP $D^+$

$Rx^n$ (R) $\Rightarrow$ A is not oxidised $\Rightarrow$ SRP $C^+$ < SRP $A^+$

$Rx^n$ (S) $\Rightarrow$ D is oxidised $\Rightarrow$ SRP $D^+$ < SRP $B^+$

SRP order $C^+ < A^+ < D^+ > B^+$

$\Rightarrow$ C is strongest reducing agent

$\Rightarrow$ B is weakest reducing agent

C > A > D > B

85)

(1) Experimentally found
(2) Due to increase in KE of molecules.

86)

Molecularity is meaning less for complex reaction.

87) $\frac{P_P^o X_P}{P_Q^o X_Q} = \frac{Y_P}{1-Y_P}$

$\frac{1}{2} \times \frac{1}{2} = \frac{Y_P}{1-Y_P}$

$Y_P = 0.2$

88)

Statement-II melting of ice is slow at high altitude due to low pressure.

89)

$E_c = (1.36 - (-0.76)) + \frac{0.06}{2} \log \frac{[Zn^{2+}][Cl^-]^2}{[Zn(s)][P_{Cl_2}]}$

$= 2.12 + 0.03 \log\left(\frac{(1)(0.5)^2}{(1)(1)}\right)$

$= 2.138$

90)

$S = \frac{K \times 1000}{\Lambda_m^o} = \frac{8 \times 10^{-5} \times 1000}{(400 \times 2)} = 10^{-4} M$ ( $\Lambda_{eq.} \times Vf = \Lambda_m$)

$K_{sp} = S^2 = 10^{-8}$

BIOLOGY

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92)

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93)

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94)

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95)

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107)

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127)

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128) NCERT, Pg. # 28

129)

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130) NCERT-XII Pg. # 6

131)

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132)

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133)

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134)

NCERT-XII, Pg. # 19, 20

135)

Hexploid wheat 2n = 42
So egg (n) = 21
embryo (2n) = 42
endosperm (3n) = 63

136)

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137)

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138) NCERT Pg # 26 (12th)

139) NEW NCERT-XII, Pg. # 38 (E), 42 (H)

140) NEW NCERT-XII, Pg. # 30 (E), 32, 33 (H)

141) NCERT Pg. # 30

142) NCERT Pg. 30

143) NCERT Pg. # 35

144)

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145)

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147)

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176) NCERT-XII, Pg. # 48

177) NCERT Pg. 48

178) NCERT XII Pg. # 45

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